Software Engineering Coursework

Contents

Aims of the system:

Features that need to be added into the system:

- Mobile app
- Registration – register their personal data (including zone, seat number and vehicles)
- Assess control systems: gated entry points (RFID tags, license plate recognition or QR code scanning
- Surveillance cameras – quick identification of unauthorised access
- Real-time monitoring and alerts – sent to security if unauthorised access attempts to enter the venue.
- Emergency monitoring – takes live locations of attendees, system provides the quickest and safest route in case of emergencies. If a lot of attendees are affected, then related emergency services could be called.
- Prevent unauthorised access
- Control undesired movements of crowds
- Quick responding to emergencies
- Safety and security are number one priority
- Improve management by registering attendees with all important credentials.

The current system downsides:

- Crowd safety and security measures must be manually implemented or managed isolated.
- No real-time safety and security features because of the huge operational overheads of the existing system.

Opposition for the implementation of the new system:

The director thinks that the app will not be widely used, he would rather develop the current system to provide real-time support rather than spending thousands on developing an app and real time monitoring system.

Representatives would rather stick to the manual system on site to deal with attendees, they believe their friendly and welcoming approach is irreplaceable. They are also reluctant to go through training with the upgraded system to deal with security and safety more effectively.

Requirement 1:

Legal, social, ethical and professional issues

The new system should be created in the public's interest as they will be its primary users. The system should understand and comply with the correct legislation, respect rules and regulations and be in the best interest of public health and safety.

Professional issues: (Employers to clients)

Employers should act professionally, with no bias or discrimination as to who registers and enters the venue. They should also respect the privacy and confidentiality of clients; this should be through the Data Protection Act 2018. This act, according to British Law, consists of:

- using information fairly, lawfully and transparently
- used for specific and explicit purposes
- used in an adequate way that is limited to what is necessary
- accurate and kept up to date
- kept no longer than necessary
- handled with appropriate security

Additionally, they should be honest about products and services, this includes not making public statements unless the individual is qualified to do so because it will stress and cause issues with attendees.

Ethical issues:

It is in the best interest of the company to create an app that will ensure we have the trust and compliance of the attendees. Firstly, we should respect the privacy of users because the system will need personal and sensitive information. If an attendee is not willing to share it then they are not able to enter the venue, this could cause issues for the safety of staff and may welcome unauthorized access if they overcome this process. The database should not share information with third parties unless we have the correct consent because if an attendee asks, we should provide all the information stored about them. Our system is based around a database therefore safe storage of information is important so if anyone with unauthorized access enters the database, they should not be able to see or obtain information. And if we need to check back on details to verify them it should be easy to find. This also ensures data integrity, to discourage from repeating data and tampering.

Legal issues:

Legal compliance must be implemented for a database system, especially because it stores data. If they follow these requirements, privacy and information security become essential to business practice. Firstly, the app should need the user's consent before obtaining any personal and sensitive information, this could be through the validation check before they must enter their details otherwise you would have to deny them access to the venue because it could cause issues other stakeholder's safety and security. Another priority is protecting attendees' information, this is to help them against spamming, phishing and robocalling. Implementing the right policies and technical reinforcements because this will be how client's information will be supported, stored, collected, used and shared. Measures are needed here to show and prevent privacy concerns and risks. Finally, the business should understand how clients are meant to interact with the business model. This is through test runs with many different stakeholders to make sure that they add the correct and necessary details when using the app.

Social issues:

Social issues must be considered to ensure the app is developed responsibly and implemented with the correct features. The main priority should be data privacy security because protecting the attendees' sensitive information is vital to prevent data misuse and breaches. We also need to implement control levels on the database system, so only the authorized people have the right to access the database, modify it and remove data. Finally, the database should be designed to minimize bias in data collection, storage and retrieval. The data validation checks should be transparent and free from discriminatory bias.

Requirement 2:

PROBLEM:

- Managing crowd – Handling huge crowds at the stadium could be difficult, making it challenging to control entry and exit, prevent unauthorized access, and manage facilities like toilets, resulting in longer queues and waiting times.
- Unauthorized access – Larger crowds increase the chance of unauthorized individuals bypassing barriers and checkpoints, whether intentionally or not. Allowing entry without a ticket reduces safety, leading to overcrowding and increases the risk of theft, vandalism, and criminal activity taking place.
- Emergency response – In a crowded environment, emergency responders struggle to reach individuals in need, delaying help, creating panic and accident risks due to crowd rushes.

PROCESS

Waterfall methodology:

The **waterfall model** is used to manage projects and develop software in a linear sequence of 6 development phases such as requirements, design, implementation, testing, deployment, maintenance. which must be completed before moving on.

Merits:

- Simple Structure – Clear, defined phases help organize the development of applications, access control, real-time monitoring system and CCTV. These complex tasks can be handled sequentially minimizing confusion and potential errors.
- Defined Requirements – Clear requirement defines the project scope, ensuring of upgrading the systems to stay within the budget £1.8 million and 18-Months timeline.
- Ease of Management – Waterfall model's structured sequential plan keeps phases on schedule, aids resource management, and simplifies project monitoring, making development smoother.

Constraints:

- Lack of flexibility – Waterfall model is rigid and does not change well to adjustments once development begins. This can be costly and time-consuming if modifications are needed later.
- Late Tests – Testing occurs only after development, causing potential errors to be discovered later, taking extra time to fix them.
- High Risk – If the apps or its features do not match the staff's requirements, user needs may be overlooked, negatively affecting the system.

V-model:

The **V-Model(Chosen Model)** process is split into verification and validation with coding in between. Verification includes requirement gathering, system analysis, software design, and module design. Validation includes acceptance, system, integration, and unit testing.

Merits:

- Early Testing – Each phase has its own testing catching errors early. Flaws will quickly be identified minimizing risks of errors later.
- Structured – Provides clear track of requirements, analysis, design, and module design ensuring thorough testing and verification.
- Quality Control – Provides high quality products by integrating the user's requirements, enhancing safety, crowd control, and monitoring system.

Constraints:

- Higher Costs – Increased testing phases increase expenses, potentially exceeding the budget £1.8 million and extending the 18–month timeline.
- Limited Flexibility – Like the waterfall model, it is rigid, mid-project changes are costly and time-consuming extending the budget and the deadline.

V-Model Software Life Cycle:

- **Gather requirements** such as user needs, emergency response protocols, and security specifications.
- In **System analysis**, design the app's interface and the system architecture.
- Develop the **Code** for the monitoring system and app. Integrate the security protocol.
- During **Verification**, perform tests for the system, unit, and integration.
- **Validation** ensures user needs are matched to product.
- **Deployment** – Execute the system
- **Maintenance** addresses any issues discovered.

Comparison:

Waterfall Methodology is linear with phases completed before the next one starts. V-Methodology is an extension of this, shaped like a V where each verification phase has matching validation tests. In waterfall model, testing only happens after the development which risks finding errors later. The V-Model catches errors earlier by testing throughout, but it is more expensive than the simpler, cheaper Waterfall model.

PEOPLE:

Stakeholders:

1) Mr. Jackson manages the budget and suggests ideas such as developing a mobile app, improving the monitoring system, and enhancing crowd management. He is eager for these features to be executed for the stadium's benefit
2) Mr. Morrison handles security, management, and crowd control, and disagrees with Mr. Jackson's new app idea due to cost concerns. He prefers upgrading the current system to save money and make it easier for staff. He plans to meet with staff to understand their needs regarding the new system.
3) End Users

- Staff can efficiently manage crowd flow, respond to emergencies, enhance security, and detect unauthorized access using monitoring system and access control system.
- Attendees will use the registration app, check in, ticket validation, and emergency guidance to help reduce crowding and ensure safety.

PROJECT:

1) The goal is to enhance safety and security amongst the crowd improve the management at the Stadium. The includes a mobile app for attendees and staff to check in via QR code. And an enhanced monitoring system for security managers to oversee surveillance, control unauthorized access, and respond to emergencies quickly.
2) List of resources:

Human resources

- Project managers oversee the overall project to be delivered withing budget and on time. They organize the correlation between technical team, stadium management and end users.
- Developers develop monitoring systems and mobile apps by creating, developing, testing, and deploying the application.
- Security staff monitor the attendees relying on monitoring and crowd management systems to track movement and respond to emergencies.
- IT support provides support by carrying out trials on the system by troubleshooting the issue ensuring the system is reliable to use.
- End users use the mobile app to verify tickets, entry, and emergency information.

Technical resources

- Software tools: Cloud service to store data, and version control system to manage changes to source code over time.
- Monitoring system: CCTVs, Access control points with QR scanners, and RFID system tracks the movement of attendees.
- Networking and Hardware Infrastructure: High speed internet service, data processing using servers to monitor crowds and data storage system to store data such as CCTV footage.

Physical resources

- Emergency equipment – First aid, and communication devices to help manage emergencies.
- Access points – A physical barrier to scan QR code for entry and verification.
- Security stations – Place for security managers to check and control access through cameras.

Breakdown of the budget

- Development costs – 50% (£900,000)

- Training costs – 25% (£450,000)
- Security equipment – 15% (£270,000)
- Maintenance costs – 10% (£180,000)

PRODUCT:

- Requirements document outlines project goals for stadium security, including app features, emergency management and monitoring crowd.
- System design document illustrates the app’s layout and integration with the security system
- Source code and executables – Development team will create the app’s main code, database schemas and APIs to connect to the stadium security. For attendees to check in executable software will be deployed.

Requirement 3:

- Definition of functional requirement: end user specifies demands in the contract, functionality must be incorporated into the system, there's an input which an operation performs on and return an expected output.
- Definition of non-functional (non-behavioral) requirements: qualities/ goals that the system must satisfy/achieve, priorities can vary from one project to another.

The functional requirements:

- A mobile app for the user registration system--> Registration must be implemented. Visitors must pre-register their personal data, zone, seat numbers and vehicle details. This data must be verified before temporary access is granted.
- Access control system--> Install gated entry points with access control systems (such as RFID tags, license plate recognition or QR code scanning) so that only authorized attendees and vehicles can enter the premises.

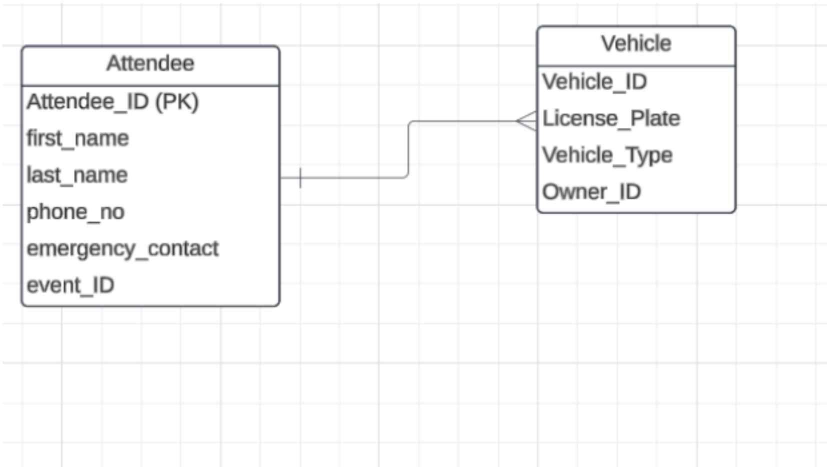
- Real time surveillance --> Surveillance cameras (such as CCTV cameras) to be used in parking spaces and entrances to identify unauthorized access.
- Real time alerts --> A centralized monitoring system should be implemented to provide real time alerts to security personnel or attendees of unauthorized people or vehicles.
- Emergency monitoring and guidance system --> System that can monitor the live location of attendees and guide them through emergency exit points
- Automated emergency notifications --> Emergency services are called (such as police, ambulance, fire brigade etc.) depending on the number of affected attendees.
- Crowd control --> The app should check the number of attendees and report to the security if there's any congestion.
- Security personnel having access to manually control the crowd --> Security personnel should be able to control measures manually if necessary.
- Attendance tracking --> A record should be kept of the entry and exit data of attendees, vehicles and staff.
- User friendly interface --> The app should be user friendly for all attendees to input data, view data and access information easily.

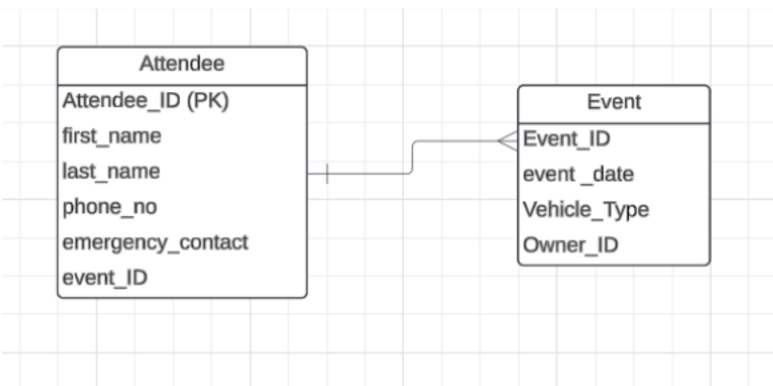
The non-functional requirements:

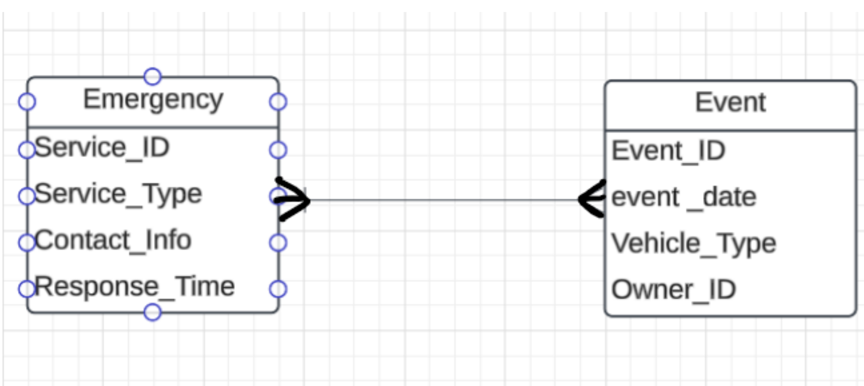
- Performance --> System needs to support real-time data and processing.
- Reliability --> System should be reliable and not crash even at a peak time of usage.
- Scalability --> System should be able to support large amounts of data due to large amounts of attendees/ events.
- Security --> Personal data of attendees and access should be stored securely to prevent misuse.
- Usability --> Interface should be easy to use for attendees and easy for staff to be trained on and help attendees with incase any problems with the app arise.

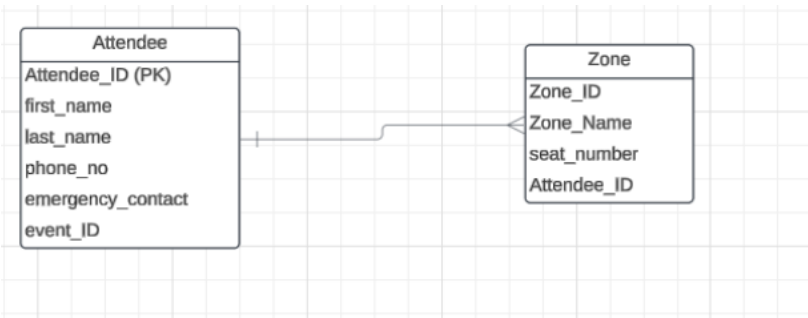
Requirement 4:

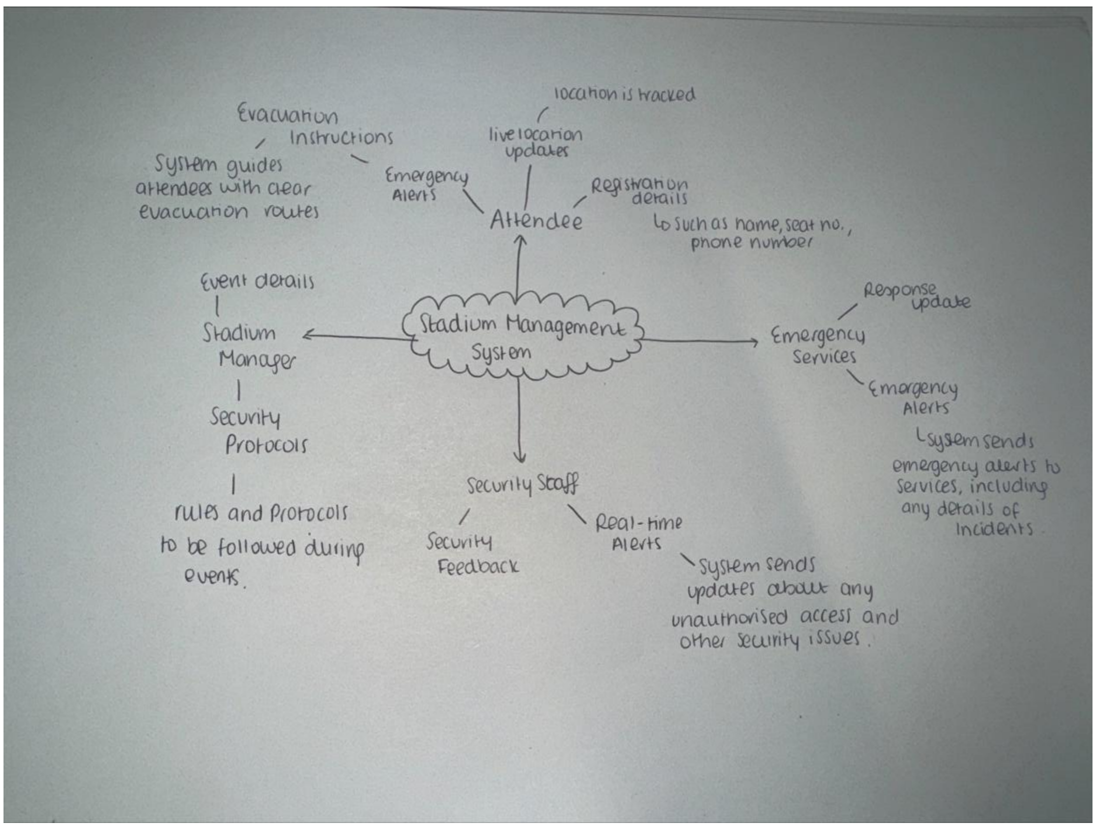
Step 1: Entity Relationship Diagram (ERD)



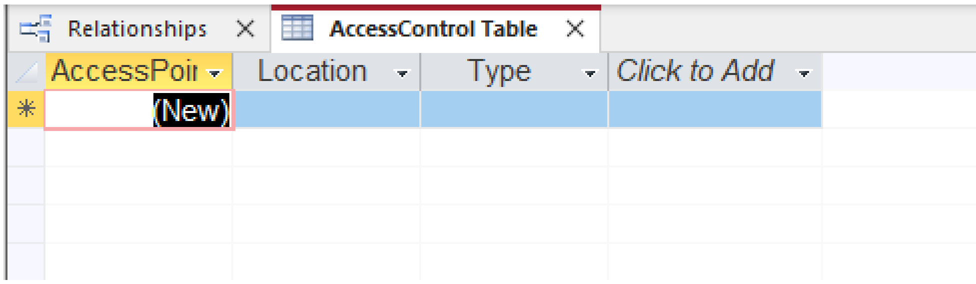


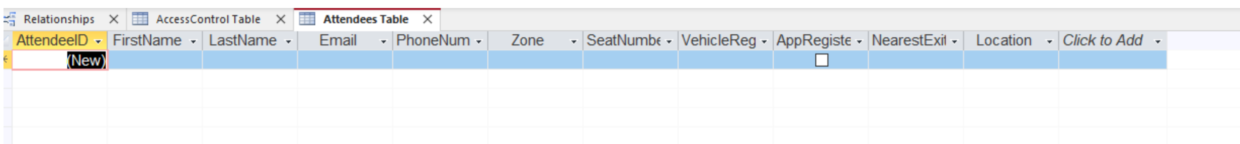


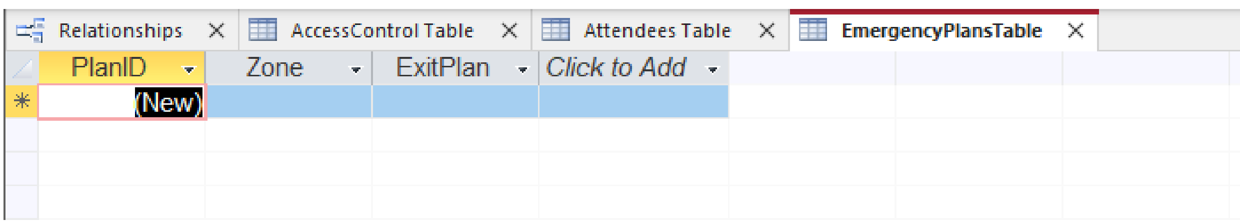

Attendee
Attendee_ID (PK)
first_name
last_name
phone_no
emergency_contact
event_ID
Zone
Zone_ID
Zone_Name
seat_number
Attendee_ID

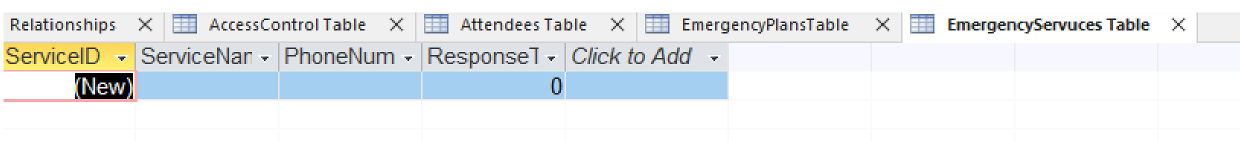

Stadium Management System
Attendee
Evacuation Instructions
System guides attendees with clear evacuation routes
Emergency Alerts
live location updates
location is tracked
Registration details
such as name, seat no., phone number
Stadium Manager
Event details
Security Protocols
rules and Protocols to be followed during events.
Emergency Services
Response update
Emergency Alerts
system sends emergency alerts to services, including any details of incidents.
Security Staff
Security Feedback
Real-time Alerts
System sends updates about any unauthorised access and other security issues.

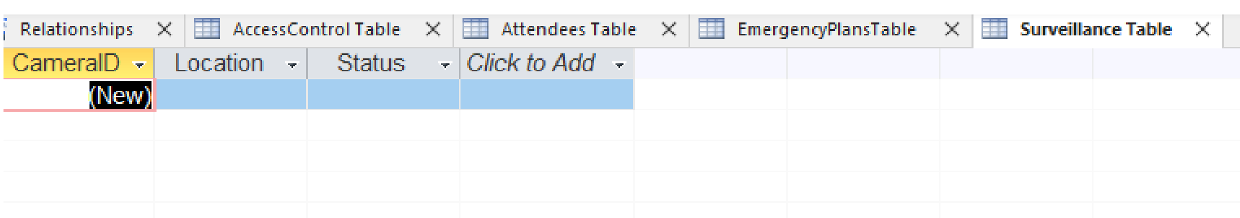
Requirement 5:











Field:	Expr1: 'Thomas'	Expr2: 'Smith'	Expr3: 'A1'	Expr4: 'South'	Expr5: 'MI54YXP'	
Table:						
Sort:						
Append To:	FirstName	LastName	SeatNumber	Zone	VehicleRegistration	
Criteria:						

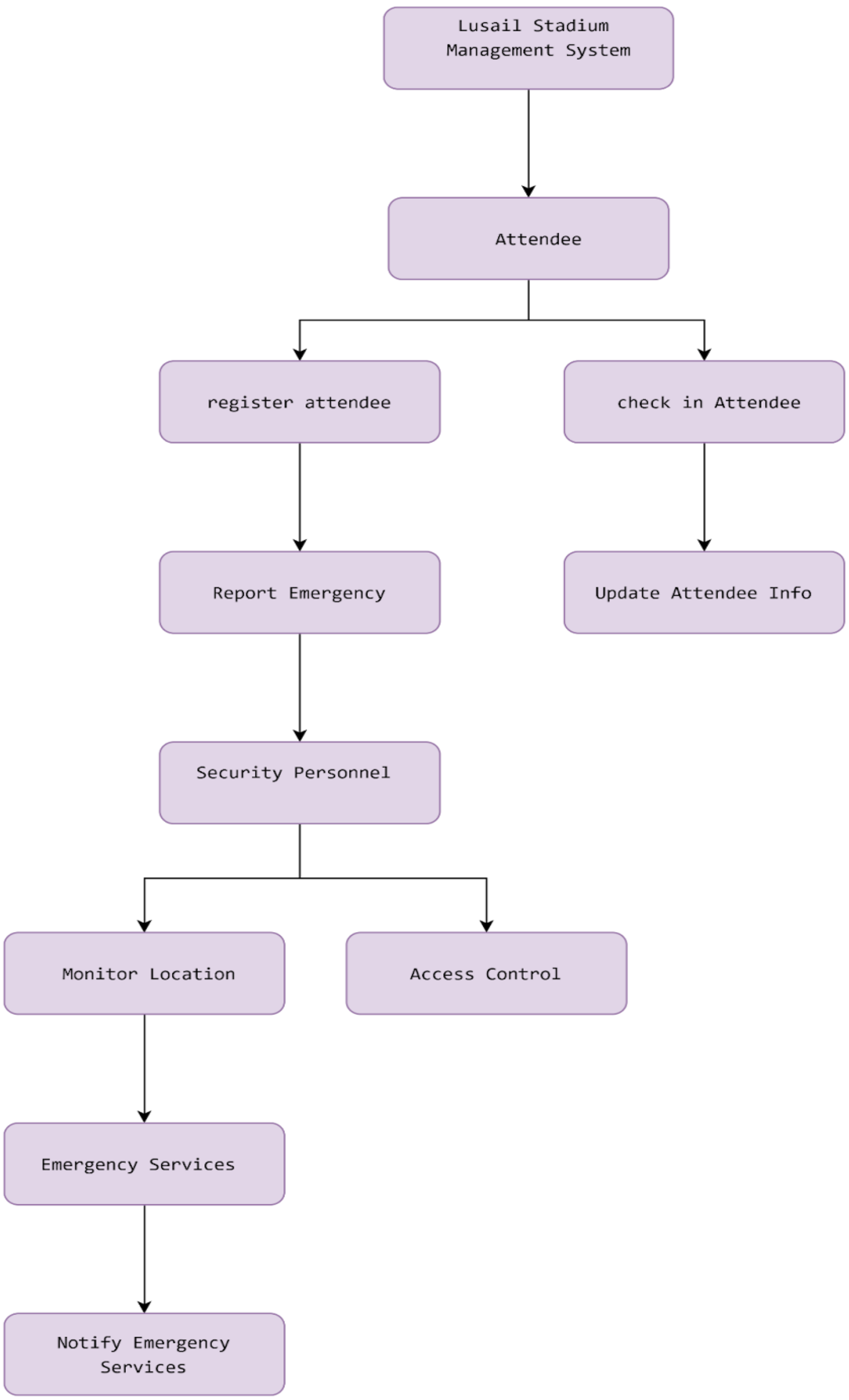
```
INSERT INTO Attendees ( FirstName, LastName, SeatNumber, [Zone], VehicleRegistration )
SELECT 'Thomas' AS Expr1, 'Smith' AS Expr2, 'A1' AS Expr3, 'South' AS Expr4, 'MI54YXP' AS Expr5;
```

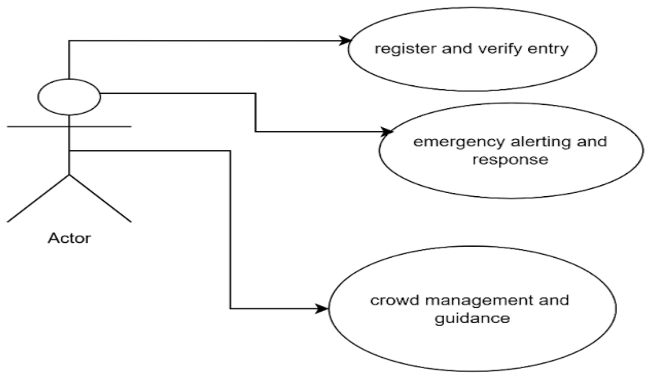
```
CREATE TABLE attendees (
    attendee_id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(250),
    seat_number VARCHAR(20),
    zone VARCHAR(20),
    vehicle_registration VARCHAR(60),
    nearest_exit_plan TEXT,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE emergency_services (
    service_id INT PRIMARY KEY AUTO_INCREMENT,
    service_type VARCHAR(50),
    contact_number VARCHAR(20),
    region VARCHAR(100)
);

CREATE TABLE locations (
    location_id INT PRIMARY KEY AUTO_INCREMENT,
    attendee_id INT,
    latitude DECIMAL(10, 6)
    longitude DECIMAL(10, 6)
    timestamp TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (attendee_id) REFERENCES attendees(attendee_id)
);

CREATE TABLE attendee_emergency_service (
    attendee_id INT,
    service_id INT,
    PRIMARY KEY (attendee_id, service_id),
    FOREIGN KEY (attendee_id) REFERENCES attendees(attendee_id),
    FOREIGN KEY (service_id) REFERENCES emergency_services(service_id)
);
```

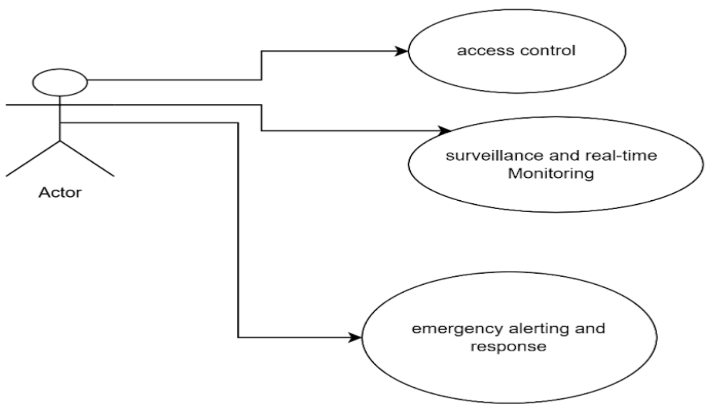
```
SELECT latitude, longitude, timestamp

FROM locations

WHERE attendee_id = 1

ORDER BY timestamp DESC

LIMIT 1;


INSERT INTO emergency_calls (attendee_id, service_id, status)

VALUES (1, 1, 'CALL PLACED');

SELECT * FROM emergency_calls

WHERE attendee_id = 1

ORDER BY call_time DESC

LIMIT 1;
```

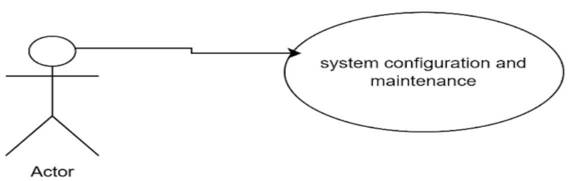
Requirement 6:

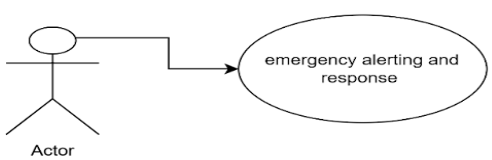
Use Case Diagram:







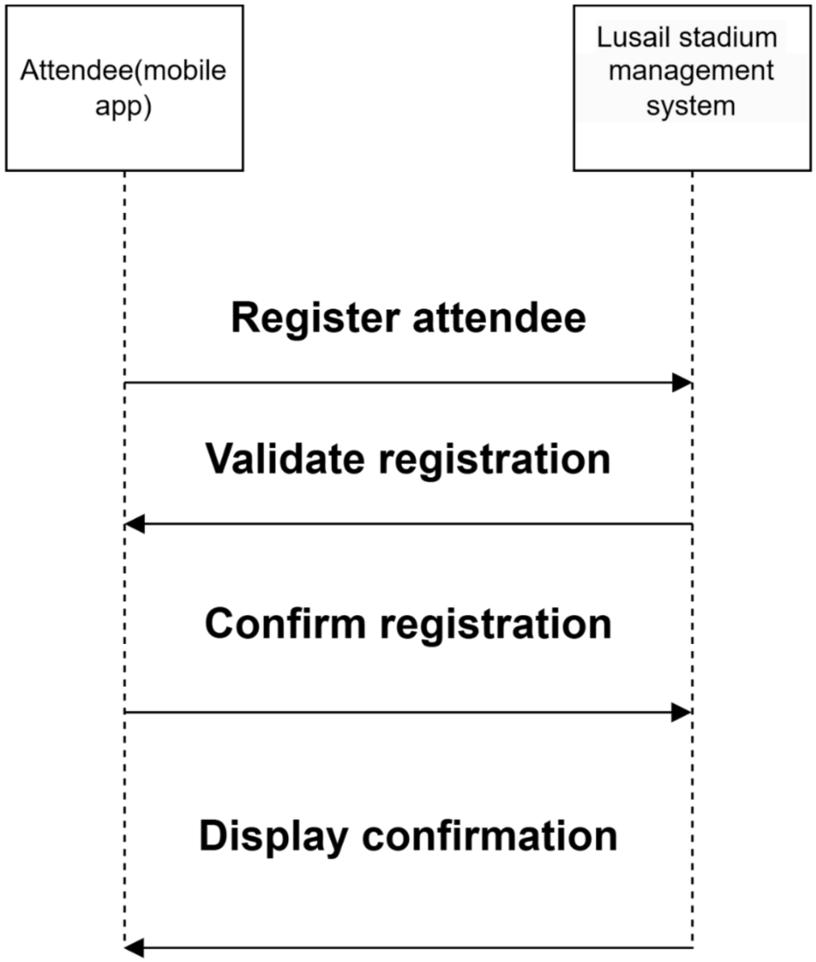




UML Sequence Diagrams

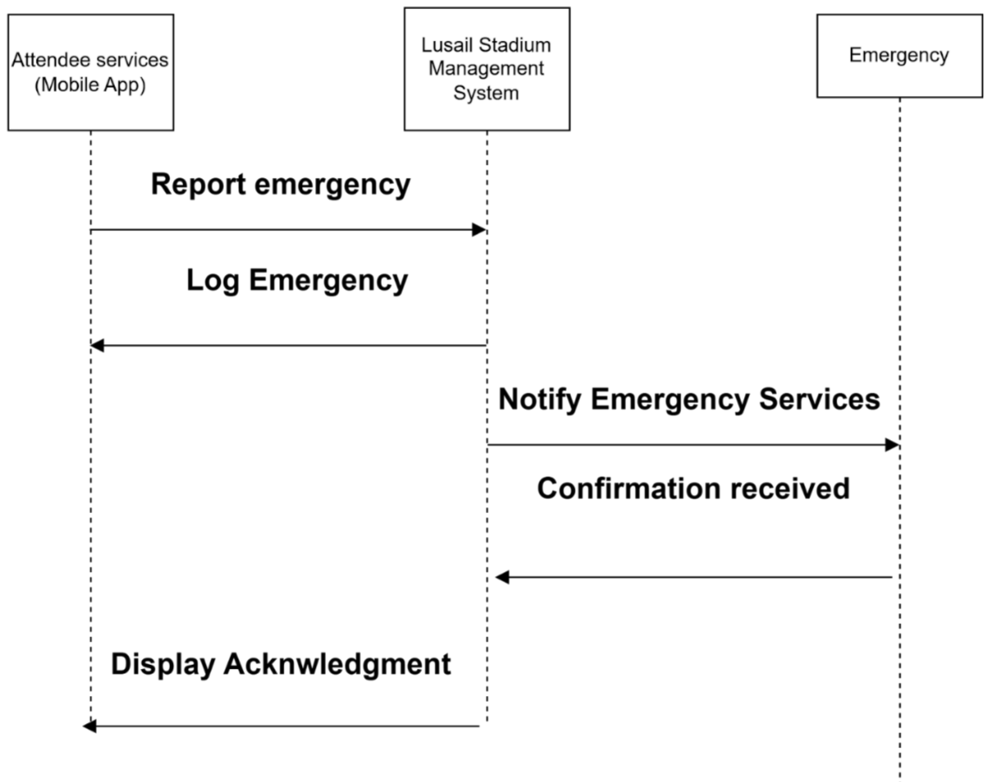
Use Case 1: Register Attendee

This sequence diagram shows the process of an attendee registering through the mobile app.



Use Case 2: Report Emergency

This sequence diagram shows how an attendee can report an emergency through the app, and how the system notifies the emergency services.



Summary:

The proposed system design leverages object-oriented analysis and design principles to create a flexible and reusable architecture. The Use Case Diagram outlines the interactions between actors and the system, while the sequence diagrams offer a detailed view of the processes involved in key use cases. This design approach ensures that the system can adapt to future changes and enhancements while supporting clarity and organization.

Proof of working together:

Task List

Search | Save Filter

New Task | E-mail List | Contact List | Show/Hide Fields | Reports | ?

Open	Task Title	Priority	Status	% Complete	Assigned To	Start Date	Due Date	Completed Date
Open	Discussion of project requirements and slide	(1) High	Completed	100%	everyone	21/10/2024	28/10/2024	27/10/2024
Open	Working on the requirements	(2) Normal	Completed	100%	everyone	28/10/2024	04/11/2024	04/11/2024
Open	Coming together to upload the best	(2) Normal	Completed	100%	everyone	04/11/2024	11/11/2024	11/11/2024
Open	Gain feedback from eachother and tutor	(2) Normal	Completed	100%	everyone	11/11/2024	18/11/2024	16/11/2024
Open	Improve work	(2) Normal	Completed	100%	everyone	16/11/2024	19/11/2024	19/11/2024
Open	Complete self assessment and slides	(1) High	Completed	100%	everyone	22/11/2024	25/11/2024	25/11/2024

Self-Assessment Sheet:

Marks mapping:

80-100 Exceptional	70-79 Excellent	60-69 Very Good	50-59 Good	40-49 Satisfactory	0-39 Fail
A	B	C	D	E	F

	A	B	C	D	E	F	Comments

the 5 Ps • Problem:			X				The 5'ps are detailed and explained amazingly, it includes everything that was described on the specification.
• Process			X				
• Project			X				

• Product			X				The model chosen should have reasons for why it was chosen for this project with advantage and disadvantage
• People			X				

functional and non-functional requirements:			x				The functional and non-functional requirements are fully accurate and relate back to the case study, it is also explained clearly.

Structured design, • Entity Relationship Diagram			X				The diagrams and queries are clear and understandable, it is easy to read, and information is given to describe what the diagrams and queries are about.
• Data Flow Diagrams.			X				
• SQL queries			X				An improvement that could be made was making additional links between the different tables in the relational diagram which could better express the one to one/one to many and many to many relationships.
			x				

UML Design Use Case Diagram - Design Class diagrams Sequence Diagrams			X X x				The diagrams are concise, comprehensive, and thorough with relevant detail.

Group members work contribution form:

Team member name	Student ID	individual overall work contribution (%)	Signature
Student1: Nabilah Akther	1378376	16.7%	NA
Student2: Ruhina Munvar	1391010	16.7%	RM
Student3: Shanjida Chowdhury	1378082	16.7%	SC
Student4: Hana Mohamadshakir	1370690	16.7%	HM
Student5: Ihsaan Eloutmani	1389932	16.7%	IE
Student6: Gizem Kumrili	1313271	16.7%	GK
		Total 100%	

Individual contributions for each task:

The contribution should indicate the effort each member contributed to a task, even if in some cases the solution of only one student was chosen for as the final one.

Describe each task you performed (For example, the creation of a Use case diagram)	Student 1: work contribution in %	Student 2: work contribution in %	Student 3: work contribution in %	Student 4: work contribution in %	Student 5: work contribution in %	total
Task: Discuss the legal, social, ethical and professional issues (LSEPI) that ought to be considered for the new system.	20%	16%	16%	16%	16%	100%
Task: the 5 Ps relevant to the given case study and document them	16%	20%	16%	16%	16%	100%

Task: A list of functional and non-functional requirements – STUDENT 6 WHICH IS BELOW	16%	16%	16%	16%	16%	100%
Task: Produce a preliminary design using traditional, SSAD, covering the full functionality of the proposed improved system	16%	16%	16%	16%	20%	100%
Task: Implement a database based on the design produced as per above requirements.	16%	16%	20%	16%	16%	100%
Task: Create the proposed system design using object-oriented analysis and design,	16%	16%	16%	20%	16%	100%

Describe each task you performed (For example, the creation of a Use case diagram)	Student 6: work contribution in %
Task: Discuss the legal, social, ethical and professional issues (LSEPI) that ought to be considered for the new system.	16%
Task: the 5 Ps relevant to the given case study and document them	16%
Task: A list of functional and non-functional requirements	20%
Task: Produce a preliminary design using traditional, SSAD, covering the full functionality of the proposed improved system	16%
Task: Implement a database based on the design produced as per above requirements.	16%
Task: Create the proposed system design using object-oriented analysis and design,	16%

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The PowerPoint:

INTRODUCTION:

Our role is to improve the stadium management for attendees.

We are creating a mobile app that allows users to:

- register their personal data
- Real-time monitoring systems and alerts
- Emergency monitoring - take live locations of attendees
- Surveillance cameras – quick identification of unauthorised access

The current system downsides:

- Crowd safety and security measures must be manually implemented or be managed isolated.
- No real-time safety and security features because of huge operational overheads of the existing system.

My role was to investigate the social, ethical, professional issues that needed to be considered for the new system. The new system should be created in the interest of the public as they will be the primary users of the system. The system should understand and comply with the correct legislation, respect rules and regulations and be in the best interest of public health and safety.

Ethical issues:
The company must develop an application that ensures trust and compliance among attendees by prioritizing user privacy. Access to personal information is necessary for entry, and refusal. It could lead to safety concerns and unauthorized access. The database must protect attendee information, requiring consent before sharing with third parties, and allowing individuals to request their data. Secure storage is essential to prevent unauthorized access, while the system should enable easy verification to maintain data integrity and prevent duplication or tampering.

Legal issues:

Legal compliance is essential for database systems to manage sensitive data, ensuring privacy and security. Applications must obtain user consent before collecting personal information, protecting all stakeholders. Safeguarding attendee data is crucial to prevent threats like spamming and phishing. Proper policies and technical measures should regulate the collection, storage, and sharing of client data. Businesses must proactively address privacy concerns and understand client interactions, which can be assessed through trial runs with stakeholders to ensure accurate information.

Social issues:
Social issues must be considered to ensure responsible application development with appropriate features. The primary concern is data privacy, as protecting participants' sensitive information is vital to prevent misuse and breaches. Control measures should be implemented to ensure only authorized individuals can access, modify, or delete data. Additionally, the database design should minimize bias in data collection, storage, and retrieval, with clear, non-discriminatory validation procedures.

Introduction:
My name is Ruhina Munvar, and today I will be presenting the 5Ps that applies to the Lusail Stadium project. This is divided into the 5 Ps Problem, Product, People, Process, and Project.

PROBLEM:

MANAGING CROWD
UNAUTHORIZES ACCESS
EMERGENCY RESPONSE

–

PRODUCT:

REQUIREMENT DOCUMENT

SYSTEM DESIGN DOCUMENT

SOURCE CODE AND EXECUTABLES

PEOPLE:

- Mr.Jackson
- Mr.Morrison
- End User

PROCESS

Methodolody		Merits	Constrains
Waterfall	It manages projects and develop software in a linear sequence of 6 development phases such as requirements, design, implementation, testing, deployment, maintenance. which must be completed before moving on.	- Simple Structure - Defined Requirements - Ease of Management	- Lack of flexibility - Late test - High risk
V-Model	This process is split into verification and validation with coding in between. Verification includes requirement gathering, system analysis, software design, and module design. Validation includes acceptance, system, integration, and unit testing.	- Early testing - Structured - Quality control	- Higher costs - Limited flexibility

Software Lifecycle - (V-Model):

1) Gather requirements
2) System analysis
3) Development of code
4) Verification
5) Validation
6) Deployment
7) Maintenance

PROJECT:

The goal is to enhance safety and security amongst the crowd improve the management at the Stadium.

List of resources:

1) Human resources
2) Technical resources
3) Physical resources

Breakdown of Budget:

1) Development costs – 50% (£900,000)
2) Training costs – 25% (£450,000)
3) Security equipment – 15% (£270,000)
4) Maintenance costs – 10% (£180,000)

3 – GIZEM

Role

We decided to use my work to focus on the third requirement: identifying the functional and non-functional requirement outlined in the case study.

Involvement within the Team

I initiated a shared Word document to display key information from the specifications for my team members. This document included checkpoints for task deadlines, feedback requests, and opportunities for overall improvement.

After our team came together to present our findings and discussed whose work is best for each requirement, we provided constructive feedback to each other and enhanced our collective work.

FUNCTIONAL REQUIREMENTS

- 1. A mobile app for the user registration system → Registration must be implemented. Visitors are required to pre-register their personal data, zone, seat numbers and vehicle details.
- 2. Access control system → Install gated entry points with access control systems (such as RFID tags, license plate recognition or QR code scanning)
- 3. Real time surveillance → Such as CCTV to control unauthorised access
- 4. Real time alerts → A centralized monitoring system should be implemented to provide real time alerts
- 5. Emergency monitoring and guidance system→System that can monitor the live location of attendees and guide them through emergency exit points
- 6. Automated emergency notifications→ Emergency services are called
- 7. Crowd control→ The app should monitor the number of attendees and report to the security if there's any congestion
- 8. Security personnel having access to manually control the crowd
- 9. Attendance tracking
- 10. User friendly interface

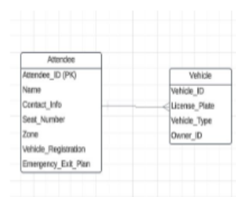
NON-FUNCTIONAL REQUIREMENTS

- 1. Performance--> System needs to support real-time data and processing.
- 2. Reliability--> System should be reliable and not crash even at a peak time of usage.
- 3. Scalability--> System should be able to maintain large amounts of data due to large amounts of attendees/ events.
- 4. Security--> Personal data of attendees and access should be stored securely to prevent misuse.
- 5. Usability--> Interface should be easy to use for attendees and easy for staff to be trained on and help attendees with incase any problems with the app arise.

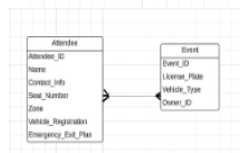
IHSSAN

ENTITY DIAGRAM AND DATA FLOW DIAGRAM

- Today I'll be presenting the Entity Relationship Diagram (ERD) and the Data Flow Diagram (DFD) for our Stadium Management System. These diagrams highlight the system's structure and how data flows through it to ensure real-time security, efficient monitoring, and quick emergency responses.



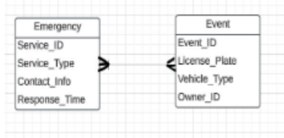
- The first relationship is between Attendee and Vehicle. This diagram shows how an Attendee is linked to their Vehicle. Each attendee will register their vehicle information in the system, which includes their vehicle's license plate, vehicle type, and entry status. This is critical for security personnel to verify authorised access to restricted areas.



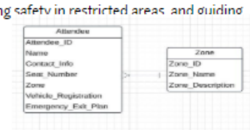
The next relationship is between the Attendee and the Event. Every Attendee is linked to an Event they are attending, with information like seat number, ticket type, and zone. This relationship helps the system monitor who is attending each event and ensures that attendees are assigned to the correct areas for their safety and comfort."

ENTITY DIAGRAM

Here, we have the relationship between the Event and Emergency services. Each Event can trigger an emergency response based on attendee data and location. If an incident occurs, the system automatically alerts the relevant emergency services, such as police or medical staff, depending on the type of emergency. This relationship ensures a quick and coordinated response to any incidents.



Lastly, we have the relationship between Attendee and Zone. Attendees are assigned to specific Zones within the stadium, which are based on their ticket type or seating preference. This helps in managing crowd flow, ensuring safety in restricted areas, and guiding attendees to their nearest exits during an emergency.



The ER diagrams demonstrate the connections between essential entities in the stadium system. These relationships support real-time management of attendees, vehicles, events, and emergencies, ensuring smooth operations and attendee safety.

DATA FLOW DIAGRAM

The central component of the system is the Stadium Management System, which interacts with several external entities like Emergency Services, Stadium Manager, Security Staff, and Attendees. This diagram outlines how data flows between the system and these key players in managing the stadium.

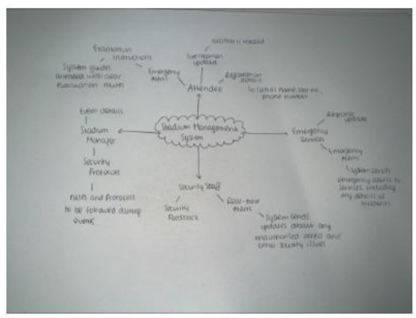
In this Data Flow Diagram, we see the Stadium Management System at the center, which serves as the core of all activities. The diagram illustrates how data moves between the system and key external entities.

"If an emergency occurs, such as a medical situation or security threat, the Stadium Management System sends alerts and detailed information to Emergency Services. The system automatically triggers relevant emergency actions based on real-time data about attendees, events, and potential threats

The Stadium Manager plays a crucial role in overseeing the entire event and stadium operations. The Stadium Management System receives data from the manager, such as event schedules, attendee information, and any specific instructions regarding crowd control or emergency preparedness.

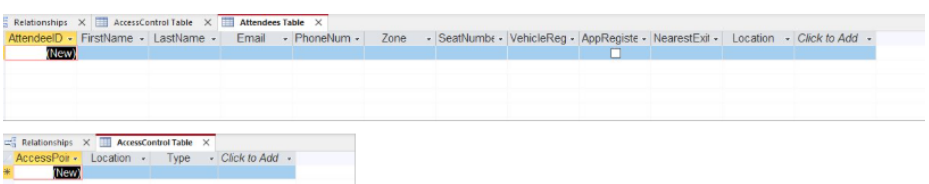
The Security Staff interacts directly with the system to monitor the security of the venue. They receive alerts from the system about any unauthorized access attempts, potential threats, or irregular activities. Security personnel use the system's data to ensure crowd safety and resolve any incidents efficiently.

Finally, the Attendees interact with the Stadium Management System primarily through their mobile app or by registering in the system. The system collects and stores their personal details, ticket information, and vehicle data. It also tracks their location within the stadium and provides real-time updates on seat assignments, emergency exits, and evacuation plans.



5- SHANJIDA

I finalised the database by creating different tables that were necessary according to the case study. There was 5 separate tables created to store different fields that will allow Lusale Stadiums to track attendees, control access ,help with emergency planning. This will be through storing LocationID and PhoneNumbers which will be implemented through a mobile app.

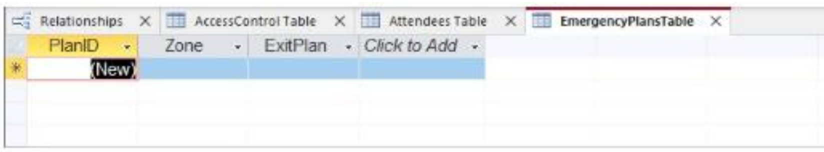


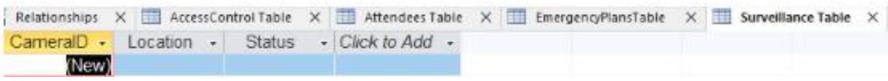
Relationships | AccessControl Table | Attendees Table

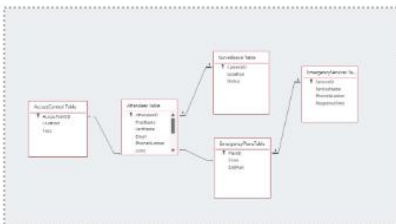
AttendeeID	FirstName	LastName	Email	PhoneNum	Zone	SeatNumbe	VehicleReg	AppRegiste	NearestExit	Location	Click to Add
(New)								☐			

Relationships | AccessControl Table

AccessPoir	Location	Type	Click to Add
(New)			

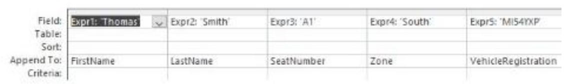




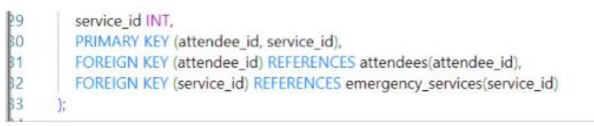


```sql
 INSERT INTO Attendees ( FirstName, LastName, SeatNumber, [Zone], VehicleRegistration )
SELECT 'Thomas' AS Expr1, 'Smith' AS Expr2, 'A1' AS Expr3, 'South' AS Expr4, 'MI54YXP' AS Expr5;

UPDATE Attendees
SET SeatNumber = 'B2', Zone = 'South
WHERE AttendeeID = 1

UPDATE VehichleRegistration
SET VehicleRegistration = 'ABC123'
WHERE AttendeeID = 1;

UPDATE ExitPlan
SET NearestExit = 'Exit 5'
WHERE AttendeeID = 1;
```

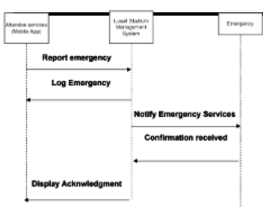
```sql
CREATE TABLE attendees (
    attendee_id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(250),
    seat_number VARCHAR(20),
    zone VARCHAR(20),
    vehicle_registration VARCHAR(60),
    nearest_exit_plan TEXT,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE emergency_services (
    service_id INT PRIMARY KEY AUTO_INCREMENT,
    service_type VARCHAR(50),
    contact_number VARCHAR(20),
    region VARCHAR(100)
);

CREATE TABLE locations (
    location_id INT PRIMARY KEY AUTO_INCREMENT,
    attendee_id INT,
    latitude DECIMAL(10, 6)
    longitude DECIMAL(10, 6)
    timestamp TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (attendee_id) REFERENCES attendees(attendee_id)
);

CREATE TABLE attendee_emergency_service (
    attendee_id INT,
```

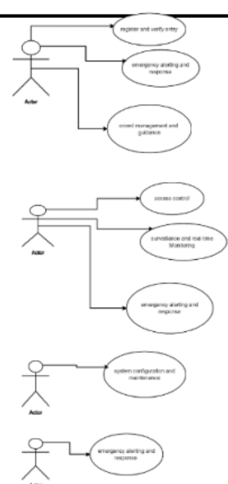


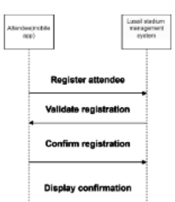
Field:	Expr1: 'Thomas'	Expr2: 'Smith'	Expr3: 'A1'	Expr4: 'South'	Expr5: 'MI54YXP'
Table:					
Sort:					
Append To:	FirstName	LastName	SeatNumber	Zone	VehicleRegistration
Criteria:					



```sql
    service_id INT,
    PRIMARY KEY (attendee_id, service_id),
    FOREIGN KEY (attendee_id) REFERENCES attendees(attendee_id),
    FOREIGN KEY (service_id) REFERENCES emergency_services(service_id)
);
```







USE CASE DIAGRAMS

Actors – entities REPRESENTATION (human & EXTERNAL)

- **Attendee**: Uses the mobile app and its functions while attending the stadium events.
- **Security Personnel**: Monitors safety, verifies credentials and controls access.
- **Emergency Services**: In case of emergencies triggered by the system, they will be called to the scene.
- **System Administrator**: Manages system operations, user permissions and account security.

Use cases – ACTIONS/TASKS PERFORMED BY ACTORS

- **Attendee**: They can register, check-in, update their profiles and report emergencies.
- **Security**: Security personnel can control access and monitor attendee movement.
- **Emergency System**: In case of any incidents, it can alert/call emergency services to come to the scene.

UML SEQUENCE DIAGRAMS

USE CASE 1: REGISTER ATTENDEE

WHAT HAPPENS WHEN AN ATTENDEE REGISTERS THROUGH THE APP

- Attendee starts registration process through app.
- They input all necessary information such as name, contact details and event details.
- The system will verify and store the submitted info in the database.
- Confirmation message/email is sent to attendee upon successfully registering.

As a result, this ensures accurate attendee records, allowing for more efficient future interactions.

USE CASE 2: REPORT EMERGENCY

WHAT HAPPENS WHEN ATTENDEE REPORTS EMERGENCIES THROUGH THE APP

- Attendee triggers emergency alert through app.
- The system logs the emergency and attendee's location.
- The system will notify emergency services with details about the incident.
- Emergency services confirm they have received the alert.
- Notifies security personnel currently in the event with details about the incident to take initial action before emergency services arrive.

As a result, this ensures that serious incidents are dealt with security and emergency services more efficiently and reliably.

SUMMARY OF DIAGRAMS' ROLES INTHE SYSTEM DESIGN

Use case diagram

Represents a high-level overview of how the system functions.

All the actors' responsibilities and the system functionalities are defined.

Sequence diagram

Visualises the step-by-step processes.

Clarifies each process's flow, ensures consistency and accuracy in the systems interactions.

SUMMARY OF DIAGRAMS ROLES IN THE SYSTEM DESIGN

Provides a broad but detailed overview of the system.

Allows for a complete understanding of the actor's responsibilities, interactions between each entity and the logical flow of critical processes within the system.